

Intro to Word Embeddings

Machine Learning

Word Embeddings

- Word Embeddings: An NLP technique of representing words as vectors
- Several methods exist to do this kind of translation. Popular methods include
 - Dimension reduction techniques, like SVD
 - word2Vec (Neural networks)
 - GloVe

Context $\approx$ Meaning

- The purpose of word embeddings is to use closer vectors to represent similar words.
- That is, it detects similarities mathematically.
- More specifically it aims to capture similarities between words based on their distributional properties in large samples of language data.
- "a word is characterized by the company it keeps"
- Given enough data, usage and contexts, word embeddings can make highly accurate guesses about a word's meaning.

Similar words and similar vectors

- Two words with similar contexts mean similar things
 - Eg: Red & Blue, Cat & Dog
- Cosine Similarity - measuring distance between word vectors
- Embedding can (surprisingly) build even more meaning into the vectors
 - Eg: 'King' - 'Man' + 'Woman' $\approx$ 'Queen'

Singular Value Decomposition

- Begins with a text corpus (Eg. All the text in wikipedia)
- Assemble a word co-occurrence matrix: M
- Find a lower dimensional word embedding matrix W such that, $M = W * W^T$.
- This step can be accomplished using Singular Value Decomposition (SVD)
- The W matrix now contains the vector representation for each word.

- I went to the cinema on Sunday
- I went to the beach on Monday
- My favorite pet is a cat
- I like walking my pet dog
- I bought a red Honda
- I bought a red car
- I bought a blue Toyota